Jesse Perla

CONTACT INFORMATION

University of British Columbia
Vancouver School of Economics
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Vancouver, BC, V6T 2E8

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http://jesseperla.com
https://github.com/jlperla/

CITIZENSHIP

Canadian, American

INTERESTS

Macroeconomics, Machine Learning, Growth, Macro-Development, Bounded Rationality

POSITIONS

Associate Professor, UBC Vancouver School of Economics
Associate Member, UBC Computer Science

OTHER AFFILIATIONS

Member, UBC Centre for Artificial Intelligence Decision-making and Action (CAIDA)
Faculty Affiliate, UBC Institute of Applied Mathematics
Core Faculty, UBC Centre for Computational Social Science
Faculty Affiliate, Stone Center for Wealth Inequality

EDUCATION

Ph.D., Economics, May 2013
New York University, New York, NY, USA
Fields: Macroeconomics, Industrial Organization
Committee: Boyan Jovanovic, Thomas J. Sargent, Jess Benhabib

B.Sc. Applied Mathematics, 1997
Columbia University, New York, NY

PUBLISHED PAPERS/REFEREED CONFERENCE PROCEEDINGS

Equilibrium Imitation and Growth (with Christopher Tonetti)
Journal of Political Economy, 2014

Catch-up and Fall-back through Innovation and Imitation (with Jess Benhabib and Christopher Tonetti)
Journal of Economic Growth, 2014

Equilibrium Technology Diffusion, Trade, and Growth (with Christopher Tonetti and Michael Waugh)
American Economic Review, 2021

Reconciling Models of Diffusion and Innovation: A Theory of the Productivity Distribution and Technology Frontier (with Jess Benhabib and Christopher Tonetti)
Econometrica, 2021

Eigendecomposition of Tau Matrices and Applications (with Sven-Erik Ekstrom, Carlo Garoni, and Adam Jozefiak)
Linear Algebra and Its Applications, 2021

Commitment and Investment Distortions Under Limited Liability (with Carolin Pflueger and Michal Szkup)
Journal of Economic Theory, 2024

STEER-ME: Assessing the Microeconomic Reasoning of Large Language Models (with N. Raman, T. Lundy, T. Amin, J. Perla, K. Leyton-Brown)
Thirty-Ninth Annual Conference on Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track, 2025.

Taming the Curse of Dimensionality: Quantitative Economics with Deep Learning (with Jesus Fernández-Villaverde and Galo Nuño)
Quantitative Economics, forthcoming

Working Papers

A Theory of Dynamic Product Awareness and Targeted Advertising (with Larent Cavenaile, Murat Celik, and Pau Roldan-Blanco)
Accepted at the Journal of Political Economy

Differentiable State-Space Models and Hamiltonian Monte Carlo Estimation (with Rachel Childers, Jesus Fernández-Villaverde, Christopher Rackauckas, and Peifan Wu)
R&R at the Journal of Econometrics

Exploiting Symmetry in High-Dimensional Dynamic Programming (with Mahdi Ebrahimi Kahou, Jesus Fernández-Villaverde, and Arnav Sood)
NBER Working Paper

Spooky Boundaries at a Distance: Inductive Bias and Dynamic Macroeconomic Models (with Mahdi Ebrahimi Kahou, Jesus Fernández-Villaverde, Sebastian Gomez, and Jan Rosa)
Conditionally Accepted at the Journal of Monetary Economics

Solving Models of Economic Dynamics with Ridgeless Kernel Regressions (with Mahdi Ebrahimi Kahou and Geoff Pleiss)
R&R at Quantitative Economics

Grants and Awards

CIFAR Catalyst Grant (2026-2028), $139,000
Economic Foundations of AI Certification
SSHRC Insight Grant (2023-2025), $155,400
Deep learning methods for solving high-dimensional equilibrium models
SSHRC Insight Development Grant (co-applicant 2021-2022), $58,635
Monopsony or Peer Effects? Disentangling the Importance of Firms in the Labor Market
Bank of Canada Governor's Award (2021-2022)

SSHRC Insight Grant (2018-2021), $159,183
Coordination frictions in macro-development: spatial equilibrium, technology traps, and industrial policy
SSHRC Insight Development Grant (2018-2020), $68,834
Debt Overhang and Over-Investment
SSHRC Insight Development Grant (2015-2017), $43,000
Why Doesn't Technology Diffuse through Migration?
Hampton Grant (2014-2015), $22,000
The Growth Dynamics of Innovation, Diffusion, and the Technology Frontier

Books

Quantitative Economics with Julia (with Thomas J. Sargent and John Stachurski)
Online textbook: https://julia.quantecon.org/

QuantEcon DataScience (with Chase Coleman, Spencer Lyon, and Collaborators)
Online textbook: https://datascience.quantecon.org/

Seminars, Conferences, and Visiting Scholar

2025: HEC Montreal, UQAM, Banff Productivity, Innovation, and Economic Growth Conference
2024: ACM-EC Workshop Organizer, ACM International Conference on AI in Finance (ICAIF)Organizer
2023: Richmond Fed, Western, Future of AI and Economics at HBS
2022: Bank of Canada, Uppsala, Peking University
2021: AEA, Canadian Macro Study Group, Bank of Canada Fellowship Learning Exchange, Boston College, Queens, Georgetown, Boston University, UCL
2020: HEC Paris
2019: SED (Organizing Committee), NBER Summer Institute (discussant)
2018: University of Pennsylvania, University of Toronto Rotman, Yale (Cowles Visiting Scholar), SED (Organizing Committee)
2017: UCSD, Stanford, Philadelphia Fed, Minneapolis Fed (Visiting Scholar), WashU Mini-Conference on Macro-Development, Minnesota Macro Theory, St. Louis Fed
2016: Simon Fraser University
2015: UC Berkeley, University of Pennsylvania, University of Pennsylvania Wharton, Columbia University, AEA, Cowles Foundation (discussant), Nottingham University Conference on Firm and Technology Dynamics

2014: San Francisco Federal Reserve, Cowles Foundation (discussant)
2013: UCLA, UCSD, Minnesota

Teaching Experience

University of British Columbia, Vancouver, BC, Canada
ECON526 - Quantitative Economics with Data Science Applications (masters)
ECON622 - Computational Economics with Data Science Applications (PhD)
Created course and moved to approval by the UBC Senate
ECON323 - Quantitative Economic Modeling with Data Science Applications (undergraduate)
Created course and moved to approval by the UBC Senate
ECON302 - Intermediate Macroeconomics (undergraduate)
ECON307 - Honors Intermediate Macroeconomics (undergraduate)
ECON408 - Computational Methods in Macroeconomics (undergraduate)
Created course and moved to approval by the UBC Senate
ECON546 - Monetary Theory and Policy (graduate)

Service

University of British Columbia, Vancouver, BC, Canada
Chair, Graduate Committee
Computer Lab Committee
Comps Committee
Executive Committee
Instructor Recruitment Committee
Recruitment Committee
Bank of Canada Governor's Challenge (Undergrad Team won 1st Place)
Centre for Innovative Data in Economics Research (CIDER) Board Member
VSE Director Selection Committee
Computer Science Recruitment Committee
Advisory Committee of the SciML (https://sciml.ai/) organization

PhD Student Placements

University of British Columbia, Vancouver, BC, Canada
Colin Caines, Committee Member (Placed at Federal Reserve Board)
João Alfredo Galindo da Fonseca, Committee Member (Placed at Université de Montréal)
Sev Hou, Advisor (Placed at Chinese University of Hong Kong, Shenzhen now SFU)
Dongxiao Zhang, Committee Member (Placed at Veeva Systems)
Nadhanael Gemaliel Vimala, Committee Member (Reserve Bank of India)
Mahdi E. Kahou, Supervisor (Placed at Bowdoin College)
Ali Karimarad, Committee Member (Placed at University of Washington)
Alejandro Roja-bernal, Committee Member (Placed at University of Hawaii)
Josh Catalan, Committee Member
Jan Rosa, Committee Member (Placed at OECD)
Giulia Lo Forte, Committee Member (Placed at Fed Board)
Mila Markevych, Committee Member (Placed at UNC Charlotte)
Neil Newman (Computer Science), Committee Member
Betty Shea (Computer Science), Committee Member
Narun Raman (Computer Science), Committee Member
Taylor Lundy (Computer Science), Committee Member
Talha Naeem, Committee Member

Referee

Econometrica, American Economic Review, Quarterly Journal of Economics, Journal of Political Economy, Review of Economic Studies, AEJ: Macro, Journal of Economic Growth, Journal of International Economics, International Economic Review, Review of Economic Dynamics, European Economic Review, Journal of Economic Theory, Theoretical Economics, Journal of Development Economics, Canadian Journal of Economics, Journal of Economic Literature, Macroeconomic Dynamics, Journal of Economic Behavior & Organization

Other Experience

CTO/Founder/Director of Research/Software Engineer at technology firms (1995-2006)